

AI/ML-Based Diabetes Prediction and Reinforcement Learning for Treatment Recommendation

U4 – Assessment 1 – CSA1717 – Artificial Intelligence

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1. Introduction

Diabetes is a major healthcare problem that requires early identification and continuous management. Machine Learning (ML) can be used to analyse patient information such as age, blood pressure, cholesterol, glucose level, BMI and family history to predict diabetes risk.

This project develops a two-stage intelligent healthcare system. First, supervised machine learning models are used to predict whether a patient is diabetic. Decision Tree and Logistic Regression are considered for diagnosis and risk stratification. Second, Reinforcement Learning (RL) is used as a conceptual framework for recommending suitable treatment actions based on changes in the patient's health state over time.

The main objectives are:

- To prepare and preprocess patient data.
- To develop an inductive learning model for diabetes prediction.
- To build and evaluate a Decision Tree classifier.
- To compare the Decision Tree with Logistic Regression.
- To identify the most important diabetes predictors.
- To model treatment recommendation as a Markov Decision Process.
- To demonstrate Q-Learning for treatment recommendation.
- To identify ethical considerations associated with AI in healthcare.

2. Dataset and Problem Definition

The healthcare company has collected patient records containing:

- Age
- Blood pressure
- Cholesterol level

- Glucose level
- BMI
- Family history

The target variable is **Diabetes Status**, represented as:

- 0 – Non-diabetic
- 1 – Diabetic

The problem can therefore be formulated as a **binary classification problem**.

The overall system consists of:

Patient Data → Data Preparation → Diabetes Prediction → Risk Stratification → Treatment Recommendation

3. Task 1: Data Preparation and Inductive Learning

3.1 Inductive Learning Approach

Inductive learning allows a machine learning model to learn general patterns from labelled examples and apply those patterns to unseen patient records.

Since the dataset contains patient features and known diabetes outcomes, **supervised classification** is appropriate.

Possible algorithms include:

- Logistic Regression
- Decision Tree
- Random Forest
- Support Vector Machine

For this project, Decision Tree and Logistic Regression are selected because they provide useful prediction performance while also allowing relatively understandable interpretation.

Target Variable

The target variable is:

Diabetes Status

The model learns the relationship between patient characteristics and diabetes status.

Important Features

Glucose Level:

Glucose is expected to be one of the strongest predictors because abnormal blood glucose is directly associated with diabetes.

BMI:

BMI can provide information about metabolic risk and is useful for identifying patients at increased risk.

Age:

Age can be associated with changing diabetes risk, particularly for type 2 diabetes.

Other relevant features include blood pressure, cholesterol and family history.

3.2 Handling Class Imbalance

A healthcare dataset may contain significantly more non-diabetic patients than diabetic patients. This is known as **class imbalance**.

Simply maximising accuracy can be misleading. For example, if most patients are non-diabetic, a model could achieve high accuracy while failing to detect diabetic patients.

Selected Method: SMOTE

SMOTE (Synthetic Minority Over-sampling Technique) can be used to address the imbalance.

SMOTE generates synthetic minority-class examples based on existing minority observations.

Advantages include:

- Improves representation of diabetic cases.
- Reduces bias toward the majority class.
- Helps the model learn minority-class patterns.

SMOTE must be applied **only to the training data after the train-test split** to prevent data leakage.

Model evaluation should focus not only on accuracy but also on:

- Precision
- Recall/Sensitivity
- F1-score
- ROC-AUC

Recall is particularly important because false negatives represent diabetic patients who were incorrectly classified as non-diabetic.

3.3 Train-Test Split and Preprocessing

The dataset is divided into:

- **70% training data**
- **30% testing data**

A **stratified 70:30 split** is preferred so that the proportion of diabetic and non-diabetic patients remains approximately similar in both datasets.

Preprocessing

Missing Values

Missing numerical values can be handled using methods such as median imputation. Categorical missing values can be handled using the mode or an appropriate category.

Normalisation/Standardisation

Numerical features can be standardised using:

$$z = (x - \mu) / \sigma$$

This places numerical features on comparable scales and is particularly useful for Logistic Regression.

Encoding

Categorical variables such as family history can be converted into numerical values:

- No $\rightarrow$ 0
- Yes $\rightarrow$ 1

One-hot encoding can be used for categorical variables containing more than two categories.

Overall Pipeline

Dataset → Cleaning → Stratified 70:30 Split → Preprocessing → SMOTE on Training Data → Model Training → Testing → Evaluation

4. Task 2: Decision Tree for Diagnosis

4.1 Decision Tree Classifier

A Decision Tree predicts diabetes by repeatedly splitting the dataset according to patient features.

The target is:

- 0 → Non-diabetic
- 1 → Diabetic

The tree can use glucose level, BMI, age, blood pressure, cholesterol and family history.

Root Node Selection

The root node can be selected using **Information Gain** or the **Gini Index**.

Information Gain is calculated as:

$$\text{Information Gain} = \text{Entropy}(\text{parent}) - \text{Weighted Entropy}(\text{children})$$

The feature with the highest Information Gain is selected.

An illustrative example is:

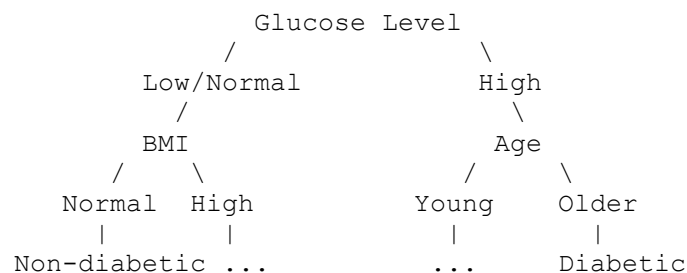
Feature	Information Gain
Glucose Level	0.25
BMI	0.12
Age	0.10
Blood Pressure	0.07

Feature	Information Gain
Cholesterol	0.04

Therefore, **Glucose Level** would be selected as the root node in this example.

First Three Levels

An example tree structure is:



The actual thresholds and branches depend on the trained dataset.

4.2 Model Evaluation

The Decision Tree is evaluated using the independent test set.

Accuracy

$$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$$

Precision

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

Recall

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

F1-Score

$$\text{F1} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

An illustrative result is:

Metric	Example Result
Accuracy	86%
Precision	82%
Recall	79%
F1-Score	80%

These values are examples and must be replaced with actual results if the dataset is available.

False Negatives

A false negative occurs when:

Actual = Diabetic

Prediction = Non-diabetic

False negatives are particularly important in healthcare because a patient with diabetes may not be identified for further clinical evaluation.

False negatives can be reduced through:

- Class weighting
- SMOTE
- Threshold tuning
- Hyperparameter optimisation
- Focusing on recall/sensitivity during model selection

4.3 Post-Pruning

Decision Trees can become excessively complex and overfit the training data.

Cost-complexity pruning introduces a parameter α that penalises overly complex trees.

The objective is approximately:

Cost = Error + $\alpha \times$ Number of Leaves

A higher α results in a simpler tree.

Pre-Pruning vs Post-Pruning

Feature	Pre-Pruning	Post-Pruning
Application	During tree construction	After tree construction
Method	Restricts tree growth	Removes unnecessary branches
Overfitting control	Good	Good
Risk	Useful splits may be stopped early	Full tree is evaluated before pruning
Complexity	Controlled during training	Reduced after training

Illustrative comparison:

Model	Accuracy
Unpruned Tree	82%
Pre-Pruned Tree	85%
Post-Pruned Tree	87%

A post-pruned tree may be preferred if it provides better generalisation while maintaining high sensitivity.

5. Task 3: Statistical Learning for Risk Stratification

5.1 Logistic Regression

Logistic Regression is selected as the statistical learning model.

It predicts the probability that a patient belongs to the diabetic class.

The output is between 0 and 1 and can be converted into a class prediction using an appropriate threshold.

The model uses predictors such as:

- Glucose level

- BMI
- Age
- Blood pressure
- Cholesterol
- Family history

5.2 Comparison with Decision Tree

Both models should be evaluated using the **same test dataset**.

An illustrative comparison is:

Model	Accuracy	AUC-ROC
Decision Tree	86%	0.84
Logistic Regression	89%	0.91

These are illustrative values and should be replaced with actual experimental results.

AUC-ROC

AUC measures how effectively a model distinguishes between diabetic and non-diabetic patients.

- **AUC = 1.0:** Perfect discrimination
- **AUC = 0.5:** Random classification
- **Higher AUC:** Better discrimination

If Logistic Regression has a higher AUC than the Decision Tree, it demonstrates stronger overall discrimination in the example.

When Statistical Models Are Preferable

Logistic Regression may be preferred when:

- Interpretability is important.
- Probability estimates are required.
- The relationship between predictors and outcome is relatively simple.
- The dataset is not extremely complex.
- Clinical professionals need to understand predictor effects.
- Computational efficiency is important.

Logistic Regression is particularly attractive in healthcare because its coefficients can provide information about the direction and strength of associations between predictors and the outcome.

6. Feature Importance Analysis

Feature importance helps identify which patient characteristics contribute most strongly to diabetes prediction.

Decision Tree

Decision Tree feature importance is based on the reduction in impurity caused by each feature.

Logistic Regression

For Logistic Regression, feature importance can be examined using the magnitude of coefficients after numerical features have been standardised.

An illustrative comparison is:

Rank Decision Tree Logistic Regression

1	Glucose Level	Glucose Level
2	BMI	BMI
3	Age	Age

Both models therefore agree on the three strongest predictors in this example.

Interpretation

Glucose Level is expected to be highly informative because it is directly associated with diabetes.

BMI provides information about metabolic risk.

Age can capture age-related changes in diabetes risk.

The models may not always agree completely because they calculate importance differently. A Decision Tree evaluates how much a feature improves tree splits, whereas Logistic Regression estimates the contribution of predictors within a statistical model.

Agreement between models increases confidence that the identified predictors contain meaningful information.

7. Task 4: Reinforcement Learning for Treatment Recommendation

7.1 MDP Formulation

Treatment recommendation can be modelled using a **Markov Decision Process (MDP)**.

The MDP consists of:

$$\text{MDP} = (\mathbf{S}, \mathbf{A}, \mathbf{P}, \mathbf{R}, \gamma)$$

where:

- $\mathbf{S}$ = States
- $\mathbf{A}$ = Actions
- $\mathbf{P}$ = Transition probabilities
- $\mathbf{R}$ = Rewards
- γ = Discount factor

States

Example patient health states are:

State	Condition
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S1	High-risk / poor health
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S2	Moderate-risk
----	---------------

S3	Improving
----	-----------

S4	Controlled / stable
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States should ideally be derived using clinically validated criteria.

Actions

The agent has four action categories:

- **Diet**
- **Exercise**
- **Medication**
- **Monitor**

Medication-related actions should not be autonomously prescribed by the AI system. They require qualified clinical supervision.

Reward

Example reward structure:

Outcome	Reward
Significant improvement	+10
Moderate improvement	+5
No significant change	0
Deterioration	-5
Serious adverse outcome	-10

The reward encourages the agent to select actions associated with positive health outcomes while penalising deterioration.

Transition Dynamics

The transition function describes how the patient can move between health states after an action.

Example:

S1 --Diet-----> S2
S1 --Exercise--> S2
S2 --Diet-----> S3
S2 --Monitor---> S2
S3 --Diet-----> S4
S3 --Exercise--> S4

In a real application, transition probabilities should be estimated using reliable longitudinal clinical data and validated clinical knowledge.

8. Q-Learning

Q-Learning learns the expected long-term reward for each state-action pair.

The Q-learning update equation is:

$$Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$$

Assume:

- Learning rate $\alpha = 0.5$
- Discount factor $\gamma = 0.9$
- Initial Q-values = 0

Five Patient Episodes

An illustrative training sequence is:

Episode	State	Action	Next State	Reward
1	S1	Diet	S2	+5
2	S1	Exercise	S2	+5
3	S2	Diet	S3	+5
4	S2	Exercise	S3	+5
5	S3	Diet	S4	+10

Iteration 1

For S1 and Diet:

$$Q(S1,Diet) = 0 + 0.5[5 + 0.9(0) - 0]$$

$$Q(S1,Diet) = 2.5$$

For S1 and Exercise:

$$Q(S1, \text{Exercise}) = 2.5$$

For S2 and Diet:

$$Q(S2, \text{Diet}) = 2.5$$

Therefore:

State Diet Exercise Medication Monitor

S1	2.5	2.5	0	0
S2	2.5	0	0	0
S3	0	0	0	0
S4	0	0	0	0

Iteration 2

Suppose:

$$\max Q(S3, a) = 5$$

For S2 and Diet:

$$Q(S2, \text{Diet}) = 2.5 + 0.5[5 + 0.9(5) - 2.5]$$

$$Q(S2, \text{Diet}) = 6.0$$

Suppose the maximum Q-value in S2 is 6.0.

For S1 and Exercise:

$$Q(S1, \text{Exercise}) = 2.5 + 0.5[5 + 0.9(6.0) - 2.5]$$

$$Q(S1, \text{Exercise}) = 6.45$$

For S3 and Diet:

$$Q(S3, \text{Diet}) = 0 + 0.5[10 + 0.9(0)]$$

$$Q(S3, \text{Diet}) = 5.0$$

Convergence Criterion

Training can be considered converged when:

$$\max |Q_{\text{new}} - Q_{\text{old}}| < 0.001$$

for a specified number of consecutive iterations/episodes.

A maximum number of episodes can also be used as a stopping condition.

9. Final Learned Policy

The final policy is obtained by selecting the action with the highest Q-value:

$$\pi(s) = \operatorname{argmax} Q(s,a)$$

An illustrative final policy is:

Patient State	Recommended Action
S1 – High Risk	Diet / Exercise
S2 – Moderate Risk	Diet
S3 – Improving	Diet
S4 – Stable	Monitor

The exact policy depends on the final Q-table obtained during training.

10. Ethical Considerations

10.1 Patient Safety

Medical decisions have significant consequences. An RL system should not independently make high-risk treatment decisions.

Medication changes should require approval from qualified healthcare professionals.

10.2 Bias and Fairness

If the training data does not adequately represent different patient groups, the system may produce biased recommendations.

The model should therefore be evaluated across relevant patient subgroups.

10.3 Explainability

Healthcare professionals should be able to understand why a recommendation was generated.

A recommendation should ideally include:

Patient State → Recommended Action → Expected Outcome → Reason

10.4 Data Privacy

Patient records contain sensitive medical information. Data should be securely stored and processed according to applicable privacy and data-protection requirements.

10.5 Human-in-the-Loop

The AI should function as a **clinical decision-support system**, rather than replacing healthcare professionals.

Doctors and other qualified professionals should be able to review and override AI recommendations.

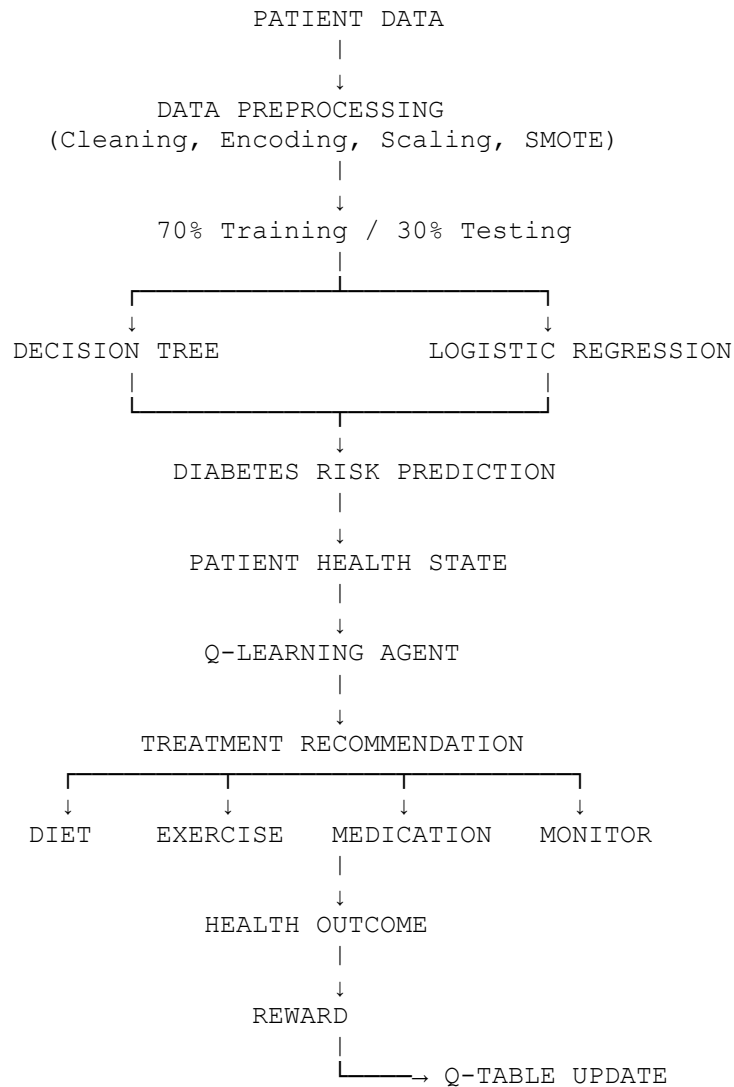
10.6 Continuous Monitoring

After deployment, the system should be continuously evaluated for:

- Performance degradation
- Bias
- Unexpected behaviour
- Adverse outcomes
- Changes in patient populations

11. Overall System Architecture

The complete proposed system can be represented as:



12. Conclusion

This project demonstrates how multiple AI and Machine Learning techniques can be combined to support early diabetes detection and personalised healthcare decision-making.

First, **inductive supervised learning** is used to learn patterns from patient records. Appropriate preprocessing, a stratified 70:30 split and SMOTE can address data quality and class imbalance issues.

A **Decision Tree** provides an interpretable classification model, while **Logistic Regression** provides a statistical approach for risk stratification. Their performance can be compared using Accuracy and AUC-ROC, while feature importance analysis identifies the most influential predictors.

The treatment recommendation problem can then be formulated as an **MDP**, where patient health conditions represent states, treatment categories represent actions, health improvement represents rewards, and changes in patient condition represent transitions. **Q-Learning** allows an agent to learn a policy that maximises long-term expected reward.

However, medical AI must be developed with strong emphasis on **patient safety, fairness, explainability, privacy and human oversight**. The proposed RL system should therefore be treated as a clinical decision-support tool rather than an autonomous replacement for healthcare professionals.

Overall, combining supervised learning for **diagnosis and risk prediction** with reinforcement learning for **sequential decision support** provides a promising framework for intelligent healthcare systems, provided that the models are rigorously validated before clinical use.